



OPEN Optimal LQG controller design for inverted pendulum systems using a comprehensive approach

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Controlling nonlinear systems, such as the inverted pendulum on a moving cart, presents a well-known challenge due to the system's nonlinearities and highly coupled states. This paper explores the control methodology of the system by linearizing the dynamics around the pendulum's upright position. The primary objective of this review article is to develop control strategies that can not only stabilize the system but also respond effectively to external disturbances. Although control techniques like Proportional-Integral-Derivative (PID), Linear Quadratic Regulator (LQR), and Model Predictive Control (MPC) are commonly used, this research places particular emphasis on the Linear Quadratic Gaussian (LQG) control method. LQG, known for its capacity to handle uncertainties and system noise, is analyzed in detail. MATLAB simulations are conducted to compare the performance of various control strategies, with a specific focus on LQG's ability to ensure stability and performance under disturbance. The findings highlight LQG's robustness in managing system uncertainties and its adaptability to changing conditions, making it a strong candidate for practical nonlinear control applications.

Keywords Inverted pendulum, Proportional-integral-derived (PID) control, Linear quadratic regulator design (LQR), Linear quadratic Gaussian control (LQG), Model predictive controller (MPC)

Control process is a field of science and engineering that explores how both natural and artificial systems behave and to subvert particular nonlinear parameters to enable the system to have a specific behavior. One of the nonlinear, inherently volatile systems, is the inverted pendulum on the cart, which inspires researchers to use various control strategies to stabilize the system. The most challenging part of the nonlinear system like moving cart inverted pendulum problem is to design the control input to nonlinear system is to stabilise the rod in the upright vertical position that to render the cartable to move while keeping the rod in the proper standing place because of its inherent instability and dynamic properties, the inverted pendulum problem is used to test and show the effectiveness of a control algorithm. The very first phase in this work is to establish the inverted pendulum equations of motion using the Euler–Lagrange equations¹.

This can handle with the process control. Process control is a constituent of a control system that changes the inputs to produce the desired output, but when unanticipated disturbances occur, the output deviates. Because of the system's uncompensated disturbance, the model's tendency to move downward toward stability, the pendulum's initial upright position, has assumed. The design of the controller will attempt to make up for this disturbance and keep the state upward. "Based on the different methodologies, a summary of the literature is provided in Table 1, which reviews various techniques used for controlling an inverted pendulum. The reference is cited in accordance with the specific method employed."

The suggested controller will attempt to account for this disruption and keep the situation upwards. There are many controlling methods, from classic controllers and artificial intelligence controller to ever. The proposed controller aims to address this disruption and maintain a positive trajectory. Various control methodologies are available, ranging from traditional controllers and artificial intelligence-driven systems to increasingly advanced and robust controllers. Classical PID controllers are adaptable and can manage system disturbances while accommodating changes in dynamics, ensuring dependable performance across diverse settings. However, they may find it challenging to sustain optimal performance in highly dynamic or nonlinear situations without further adjustments. Tuning PID parameters often depends on trial and error, a process that can be lengthy and may not produce the best outcomes without systematic strategies¹³. LQR offers an optimal control solution by minimizing a cost function that harmonizes system performance with control effort, relying on an optimal state feedback control strategy. It can address intricate designs that require precise modeling. It shows resilience to model uncertainties and can manage nonlinearity in proximity to linear operating conditions. The selection of Q and R matrices, determines the optimum value. The LQG controller can handle this inadequacy of LQR¹⁴. The

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References	Methodology used	Details of paper contribution
2	Sliding Mode Controller	It demonstrate and compare two nonlinear control strategies for driving and controlling an inverted pendulum utilising a novel actuator with two response wheels. It has used to test new controllers, such as tracking the intended trajectory rather than controlling the pendulum in the inverted state with swing-up control techniques
3	Fuzzy LQR and Fuzzy LQG	FLQR and FLQG controllers were developed specifically for DLRIP stability control and were subsequently contrasted with LQR and LQG controllers. The comparison aimed to assess their respective performance. Through testing, their ability to withstand both internal and external disturbances was evaluated, providing insights into their resilience
4	Indirect Adaptive Learning	To enhance the controller i.e. P-type from set points
5	LQG Optimal controller	The inverted pendulum is stabilised, and process and measurement noise is removed
6	LQR	To linearize the system about the vertical position, Tayler's series approximation is utilized. In a vertically straight situation, LQG is employed to address the servo problem and stabilize the angle of the pendulum rod with noise rejection
7	Active disturbance Rejection SMC	To improve the performance of control loop system while performing the stability
8	Two-loop proportional–integral–derivative (PID) controller and LQR design	The desired positions for the dominant closed-loop poles are determined through a Linear Quadratic Regulator (LQR) design within a two-loop Proportional-Integral-Derivative (PID) controller system for an inverted cart-pendulum setup
9	(FSUC) and (FC-PDPA)	A Takagi–Sugeno (T–S) fuzzy model serves as an effective approximation for a hybrid fuzzy controller within a short range of angles around its equilibrium state. This hybrid controller comprises a Fuzzy Swing-Up Controller (FSUC) and a Fuzzy Controller utilizing the Parallel Distributed Pole Assignment method (FC-PDPA)
10	Adaptive control	The described control system incorporates several key components: a stabilization regulator (LQR) for maintaining stability, a swing-up control method for energy optimization, a switching condition for transitioning between control modes, and an automated self-tuning mechanism to address parametric uncertainty
11	GWO algorithm Integrated with LQR controller	Control over two parameters is essential to effectively guide a pendulum to a predetermined point: the vertical pendulum angle and the cart position. In this context, a position controller is developed utilizing a Linear Quadratic Regulator (LQR) designed and optimized for an inverted pendulum system
12	PID control & LQR	The nonlinear dynamical inverted pendulum system was subjected to control using both the Linear Quadratic Regulator (LQR), an optimal control approach, and PID control, which are commonly employed for governing linear dynamical systems

Table 1. Review of techniques used for controlling inverted pendulum.

LQG controller combines LQR for optimal control with a Kalman filter for optimal state estimation, making it effective in handling both control tasks and noisy environments. When not all states are measurable, LQG can estimate the required states from data, reducing the need for additional sensing hardware. LQG provides robust performance, compensating for system noise and measurement errors, improving the controller's applicability in practical situations. As a linear controller, LQG may not perform well for nonlinear systems⁵, especially when the system experiences large deviations from the equilibrium point. Integrating the Kalman filter with the LQR controller increases the complexity of the system, requiring more computational resources and a detailed tuning process.

Model Predictive Control (MPC) optimizes future control actions by solving a multi-step optimization problem at each time interval, resulting in enhanced performance in dynamic scenarios¹⁵. MPC is adept at handling operational constraints, such as limits on system states and inputs, ensuring compliance with safety and operational requirements. MPC can simultaneously control multiple inputs and outputs, making it effective for complex systems with interrelated.

The organisation of this paper begins with an exposition of the system description and modelling in section “System description(Lagrange’s method)”. section “Methodology used” delineates the method employed, encompassing PID control and LQR, LQG, UKF With LQG and MPC controllers. The evaluation of performing these methodologies as has expounded upon in section “Implementation of the methodology to the system”, which is further explained through the analysis of various strategies and case studies. Finally, the findings and implications of the study has summarised in section “Conclusion”. Research has primarily focused on addressing two-dimensional scenarios, wherein the pendulum undergoes motion within the vertical plane, as depicted in Fig. 1¹⁶.

System description(Lagrange’s method)

$$\text{Kinetic energy of cart, } \frac{1}{2} M \dot{x}^2 \quad (1)$$

where, M = cart’s mass. x = cart position.

The centre of gravity (pendulum) is x_p, y_p

$$x_p = x + l \sin \theta \quad (2)$$

$$y_p = l \cos \theta \quad (3)$$

Differentiating above equation we get,

$$\dot{x}_p = \dot{x} + l \dot{\theta} \cos \theta \quad (4)$$

$$\dot{y}_p = -l \dot{\theta} \sin \theta \quad (5)$$

$$\text{Total kinetic energy (T)} = \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m (\dot{x}_p^2 + \dot{y}_p^2) \quad (6)$$

After solving the equation we get,

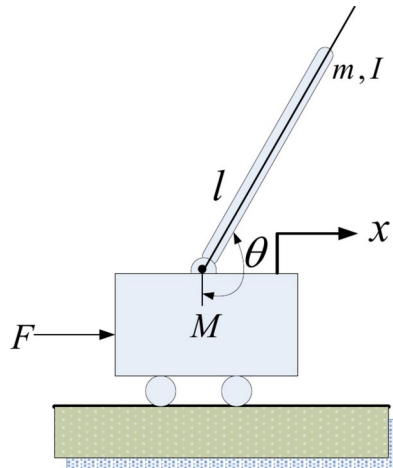


Fig. 1. Cart-inverted pendulum system.

$$\frac{1}{2} (M + m) \dot{x}^2 + m \dot{x} l \dot{\theta} \cos \theta + m l^2 \dot{\theta}^2 \quad (7)$$

Linearization of non linear system using Lagrangian Method

Lagrangian equation (L) = T - V

$$\text{Potential energy of pendulum (V)} = mgl \cos \theta \quad (8)$$

$$\text{Then, } L = \frac{1}{2} (M + m) \dot{x}^2 + m \dot{x} l \dot{\theta} \cos \theta + m l^2 \dot{\theta}^2 - mgl \cos \theta \quad (9)$$

We know that,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = 0 \quad (10)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0 \quad (11)$$

Putting the value of Eq. (9) in Eqs. (10 and 11) we get,

$$(M + m) \ddot{x} + ml \ddot{\theta} \cos \theta - ml \dot{\theta}^2 \sin \theta = 0 \quad (12)$$

$$ml \ddot{x} \cos \theta + ml^2 \ddot{\theta} - mgl \sin \theta = 0 \quad (13)$$

Assume,

$$\cos \theta \approx 1 \quad (14)$$

$$\sin \theta \approx \theta \quad (15)$$

$$\dot{\theta}^2 = 0 \quad (16)$$

After linearization we get the value

$$\ddot{x} = \frac{F - b\dot{x} - ml\ddot{\theta}}{(M + m)} \quad (17)$$

$$\ddot{\theta} = \frac{g\theta - \ddot{x}}{l} \quad (18)$$

Putting all the value of $\ddot{\theta}$ and $\ddot{x}$ we get,

$$\ddot{x} = \frac{F - b\dot{x} - mg\theta}{M} \quad (19)$$

$$\ddot{\theta} = \frac{(M + m)g\theta - F + b\dot{x}}{Ml} \quad (20)$$

State space equation,

$$x_1 = \theta \quad (21)$$

$$\dot{x}_1 = \dot{\theta} = x_2 \quad (22)$$

$$x_3 = x \quad (23)$$

$$\dot{x}_3 = \dot{x} = x_4 \quad (24)$$

$$\dot{x}_2 = \dot{\theta} = x_5 \quad (25)$$

$$\dot{x}_4 = \dot{x} = x_6 \quad (26)$$

Let output be,

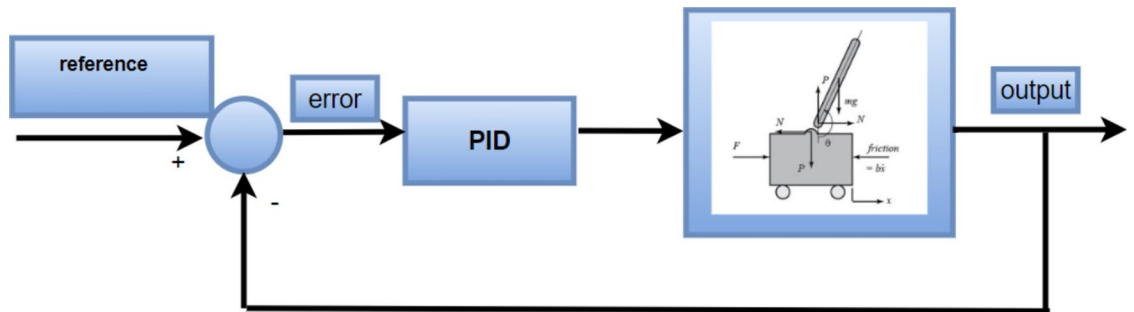
$$y_1 = \theta = x_1 \quad (27)$$

$$y_2 = x = x_3 \quad (28)$$

State space model in Matrix form.

It is represented as

Description	Value	Unit
Mass of cart “ M_c ”	0.455	Kg
Mass of pendulum “ m_p ”	0.21	Kg
Friction coefficient	0.12	$\text{Nm}^{-1} \text{s}^{-1}$
Acceleration due to gravity	9.81	m/s^2
Length of rod	0.33	m

Table 2. System parameter.**Fig. 2.** Control block diagram for Inverted pendulum using PID.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{(M+m)g}{Ml} & 0 & 0 & \frac{b}{ml} \\ 0 & 0 & 0 & \frac{1}{M} \\ -\frac{mg}{M} & 0 & 0 & -\frac{b}{M} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{1}{Ml} \\ 0 \\ \frac{1}{M} \end{bmatrix} [F] \quad (29)$$

And output equation is

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + [0] [F] \quad (30)$$

The parameter that is used in this Inverted Pendulum is shown in Table 2¹⁶:

Methodology used

This research is primarily dedicated to addressing the inverted pendulum control problem, aiming to create effective control strategies with fast response that can handle the environmental condition. Proportional-Integral-Derivative (PID) Control, Linear Quadratic Regulator Design (LQR), Linear Quadratic Gaussian (LQG), Model Predictive Controller (MPC) are a widely adopted control method that holds substantial appeal in both academic and industrial context^{17,18}. Its intuitive principles, its capability to manage multivariable systems, and its explicit consideration of equipment and state constraints underpinned its popularity. For a more comprehensive understanding of this application has encouraged to explore the comprehensive surveys provided in References¹⁹. The system can be described using Newton's laws or Lagrangian mechanics. The resulting equations of motion are nonlinear differential equations that govern the dynamics of the cart-pendulum system. The nonlinear equations for the system are generally linearized around the equilibrium point with assumption that the cart moves on a frictionless surface, the pendulum is rigid and of a certain mass and length and the control input is a force applied horizontally to the cart.

Proportional-Integral-Derivative (PID) control

The PID algorithm continues to maintain its prominence as the preferred approach for industrial process control, even in the face of ongoing advancements in control theory. This enduring popularity can attributed not only to the PID algorithm's simple and intuitively understandable structure but also to its ability to deliver satisfactory performance in a wide range of implementation^{20,21}. Control block diagram for IP using PID is shown in Fig. 2¹⁴.

$$\begin{aligned} PID(s) &= \frac{H(s)}{W(s)} = K_p + \frac{K_i}{s} + K_d s \\ &= K_P \left(1 + \frac{1}{T_i s} + T_d s \right) \end{aligned} \quad (31)$$

where $H(s)$ is control signal acting on error signal $W(s)$, K_p is proportionality gain, K_i is integral-gain, K_d is derivative—gain and T_i is integral time constant, T_d is derivative time constant.

Linear Quadratic Regulator design (LQR)

The aim is to plan a control strategy that limits the quadratic cost function associated with the state variables and control inputs. This can be accomplished using methodologies like the Linear Quadratic Regulator (LQR) or other robust control tools^{21,22}. The LQR design optimises the control inputs based on the system dynamics and desired performance requirements, and the Fig. 3 is the control block diagram of LQR Controller¹⁶.

The input that is controlling the controller is stated as:

$$u(t) = -R^{-1}B^T Px(t) - Ku(t) \quad (32)$$

The quadratic objective function for an infinite time is as $J = \int_0^{\infty} [x(t)Qx(t) + u(t)Ru(t)] dt$.

Where, $u(t)Ru(t)$ is control input in the form of quadratic in which R is a cost matrix, $x(t)Qx(t)$ as transient energy and Q acts as state weighted matrix, Q and R are matrices (diagonal), and the quadratic objective function, J is used to solve feedback gain matrix (K) then equation becomes $\dot{x} = [A - BK]x(t)$. In the designing of the LQR (Linear Quadratic Regulator) controller, certain properties and conditions must be satisfied for the weighted matrices Q and R . The Riccati equation comes from applying the Hamilton–Jacobi–Bellman (HJB) equation, which is a fundamental equation in optimal control theory. The HJB equation is generally nonlinear, but for the special case of linear systems with quadratic cost functions (as in LQR), the HJB equation simplifies to the Algebraic Riccati Equation. Thus, solving the Riccati equation yields the matrix P which is used to compute the optimal feedback gain K . The Algebraic Riccati Equation is used in LQR because it provides the optimal feedback gain K that minimizes the specified cost function for a linear system.

The cost function is simplified as

$$J = \int_0^{\infty} x(t)Qx(t) + u(t)Ru(t)dt \quad (33)$$

$$Ru + B^T\lambda = 0 \quad (34)$$

$$\lambda = Px \quad (35)$$

$$PAx + A^T Px + Qx - PBR^{-1}B^T Px + P = 0 \quad (36)$$

$$PA + A^T P + Q - PBR^{-1}B^T P + P = 0 \quad (37)$$

The equation is known as an algebraic equation of Riccati

$$u(t) = -R^{-1}B^T Px = -Kx \quad (38)$$

Minimizing the cost function, or performance index, is essential in control systems to achieve optimal system performance. It allows the system to maintain stability while minimizing deviations from the desired state and reducing control effort. By optimizing this function, we ensure a balanced trade-off between competing objectives, such as performance, energy efficiency, and response time, leading to an efficient and stable operation. Simulation results for varying values of R in the LQR controller are provided to facilitate a comparative analysis of system performance. The value of R directly impacts the control effort, revealing the trade-offs between control energy and stability.

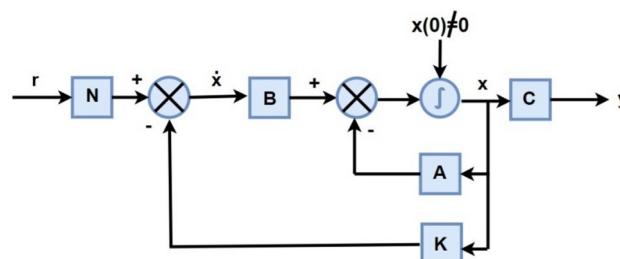


Fig. 3. Control block diagram for LQR controller.

Linear Quadratic Gaussian design for inverted pendulum on a cart

The aim is to plan a control strategy, that limits the quadratic cost function associated with the state variables and control inputs. The Linear Quadratic Gaussian (LQG) controller is created through integrating of a Kalman filter and a LQR controller for optimal control²³. This approach is based on the separation principle, which states that the optimal control and estimation problems can be solved independently, and their solutions can combine²⁴. The primary tool used in this context is the Kalman filter; specifically the Linear Quadratic Estimator (LQE) or Kalman-Bucy filter. The Kalman filter is a state estimator that uses a combination of predicted state information from the system model and measurements from sensors to estimate the true states of the system. The Kalman filter provides an optimal estimation framework for linear systems with Gaussian noise. By iteratively predicting and updating the state estimate, it minimizes the MSE, making it the most accurate linear estimator possible under these conditions. It considers the covariance matrices of process-noise and measurement-noise to optimize the estimation process and provide accurate state estimates^{25,26}.

Kalman Filter design and state estimation of Kalman filter

The Kalman filter is proven to be the optimal estimator for the state of a linear system subject to Gaussian noise. Through its recursive process of predicting and updating, it minimizes the mean squared error, making it the most effective linear estimator under these conditions.

Step 1: System Setup and Problem Formulation:

Consider a system described by the following stochastic equations:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) + w(t) \\ y(t) &= Cx(t) + v(t)\end{aligned}\quad (38)$$

where:

$x(t)$ denotes the state vector of system. $u(t)$ is the control input applied to system. $y(t)$ represents the system output measured with noise, $w(t)$, $v(t)$ are Gaussian white noise processes representing process and measurement noise with covariance matrices W and V respectively.

The objective of the Kalman filter is to estimate $x(t)$ based on $y(t)$ that minimizes the expected estimation error.

Step 2: Kalman filter estimation process:

The Kalman filter's estimate $\hat{x}(t)$ of the state $x(t)$ evolves according to:

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + K(y(t) - C\hat{x}(t)) \quad (40)$$

where K is Kalman gain to minimize the estimation error.

Define the estimation error as:

$$e(t) = x(t) - \hat{x}(t) \quad (41)$$

Differentiating $e(t)$ with respect to time gives:

$$\dot{e}(t) = \dot{x}(t) - \dot{\hat{x}}(t) \quad (42)$$

Substituting the dynamics of $x(t)$ and $\hat{x}(t)$ we find $\dot{e}(t)$

$$\dot{e}(t) = (A - KC)e(t) + w(t) - Kv(t) \quad (43)$$

This equation describes how the estimation error evolves over time.

Step 3: Estimation Error Covariance matrix:

$$\text{The estimation error covariance matrix } P = E[e(t)e(t)^T] \quad (44)$$

describes the spread or variance of the estimation error. To find $\dot{P}$, we differentiate P with respect to time. Using the product rule for differentiation

$$\dot{P} = E[e(t)e(t)^T] + E[e(t)\dot{e}(t)^T] \quad (45)$$

Substitute $\dot{e}(t)$ into expression to expand $\dot{P}$

$$\dot{P} = (A - KC)P + P(A - KC)^T + W + KVK^T \quad (46)$$

This differential equation for P is known as the Riccati differential equation.

Step 4: Optimization of the Kalman Gain K :

The Kalman filter chooses K to minimize the steady-state value of P , under the assumption that the system reaches a steady-state solution where $\dot{P} = 0$

$$AP + PA^T - PC^T V^{-1} CP + W = 0 \quad (47)$$

To solve for K , we use:

$$K = PC^T V^{-1} \quad (48)$$

The result is an optimal steady-state Kalman filter that minimizes the estimation error covariance for the given system.

Disturbance rejection in LQG

Disturbance rejection is a critical aspect of control systems, particularly in scenarios where external factors can influence the system's dynamics. In this research paper, we focus on applying Linear Quadratic Gaussian (LQG) control, which integrates the Kalman filter for state estimation and the Linear Quadratic Regulator (LQR) for optimal control. Through this integration, LQG control offers a practical approach to minimize the impact of disturbances on the controlled system^{17,27}. This paper explores the mechanism of disturbance rejection in LQG control, emphasizing the synergy between state estimation, optimal control, and feedback mechanisms. By leveraging the Kalman filter to estimate the state and disturbances affecting the system, the LQR controller adjusts the control input to mitigate the effects of disturbances and is shown in Fig. 4²⁸.

$$s = Hp + Kw + Kd \quad (49)$$

$$o = pY \quad (50)$$

Integrator output is denoted by $z(t)$ and the system is

$$y = i - o \quad (51)$$

$$y = i - pY \quad (52)$$

$$\tilde{x} = [p \ z]^T \quad (53)$$

$$\begin{bmatrix} s \\ y \end{bmatrix} = \begin{bmatrix} H & 0 \\ -y & 0 \end{bmatrix} \begin{bmatrix} p \\ y \end{bmatrix} + \begin{bmatrix} K \\ 0 \end{bmatrix} v + \begin{bmatrix} 0 \\ 1 \end{bmatrix} i + \begin{bmatrix} K \\ 0 \end{bmatrix} v \quad (54)$$

$$v = [B \ N] \begin{bmatrix} p \\ y \end{bmatrix} \quad (55)$$

$$\begin{bmatrix} s \\ y \end{bmatrix} = \begin{bmatrix} H + KB & KN \\ -Y & 0 \end{bmatrix} \begin{bmatrix} p \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} i + \begin{bmatrix} K \\ 0 \end{bmatrix} d \quad (56)$$

$$o = [Y \ 0] \begin{bmatrix} p \\ y \end{bmatrix} \quad (57)$$

$$\Delta_g(S) = \det \begin{bmatrix} SI - H - KB & -KN \\ Y & S \end{bmatrix} \quad (58)$$

Above equation is the equation of key.

The transfer function from u to o is

$$\frac{o(s)}{u(s)} = Y(SI - H - KB)^{-1} \cdot B \quad (59)$$

After getting the determinant of above equation,

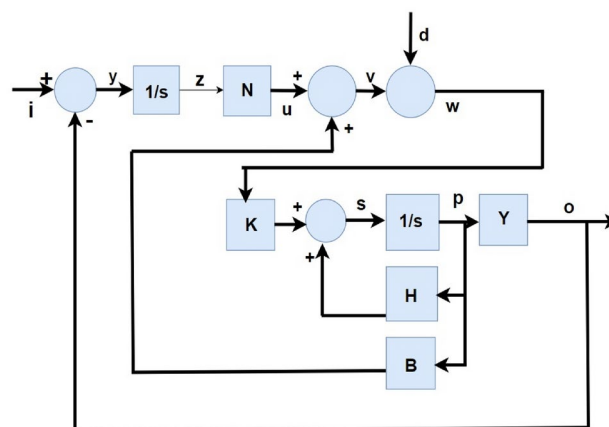


Fig. 4. Schematic diagram with disturbance input.

$$1. \Delta_g(S) = \bar{A}(S)(S + \frac{\bar{C}(S)}{\bar{A}(S)}) \cdot N \quad (60)$$

$$\Delta_g(S) = S\bar{A}(S) + N\bar{C}(S) \quad (61)$$

$$\hat{g}_{od}(S) = \frac{S\bar{C}(S)}{S\bar{A}(S) + N\bar{C}(S)} = \frac{S\bar{C}(S)}{\Delta_g(S)} \quad (62)$$

$$\hat{g}_{oi}(S) = \frac{N\bar{C}(S)}{S\bar{A}(S) + N\bar{C}(S)} = \frac{N\bar{C}(S)}{\Delta_g(S)} \quad (63)$$

In practical implementations of LQG control, several factors must be considered to ensure effective disturbance rejection. These include the accuracy of the system model used in the Kalman filter, the selection of appropriate weighting matrices for the cost function in the LQR controller, and the sampling rate of sensor measurements and control inputs. Moreover, robustness to modeling errors, sensor noise, and uncertainties in disturbances should be addressed through appropriate design and tuning of the controller parameters.

Unscented Kalman Filter (UKF) Integrated with Linear Quadratic Gaussian (LQG) control

When system dynamics are nonlinear, the traditional Kalman filter may fail. For nonlinear systems like the inverted pendulum, the Unscented Kalman Filter (UKF) is an improvement over the standard Kalman filter. It approximates the state distribution by propagating sigma points through the nonlinear dynamics rather than relying on linear approximations. This derivation focuses on combining the UKF for state estimation with the LQR for control within the LQG framework²⁹.

The Linear Quadratic Gaussian (LQG) controller combines two essential control strategies:

- **Linear Quadratic Regulator (LQR):** Optimizes the control input by minimizing a quadratic cost function.
- **Kalman Filter (KF):** Provides optimal state estimation under Gaussian noise.

Design of Unscented Kalman Filter (UKF) for nonlinear state estimation for inverted pendulum

The inverted pendulum on a cart can be described by a state-space representation with nonlinear dynamics:

$$\dot{x}(t) = f(x(t), u(t)) + w(t), z(t) = h(x(t)) + v(t) \quad (64)$$

where, $x(t) = [x, \dot{x}, \theta, \dot{\theta}]^T$ is the State vector.

x = Cart position, $\dot{x}$ = Cart velocity, θ = Pendulum angle, $\dot{\theta}$ = Pendulum angular velocity.

$u(t)$ is the control input, $w(t)$ is the process noise, $v(t)$ is the measurement noise, $z(t)$ is the measurement vector, typically cart position and pendulum angle.

The UKF estimates the state by capturing the mean and covariance of a set of sigma points that approximate the probability distribution of the state.

Step 1: Sigma point generation

A set of $2n + 1$ sigma points χ_i are generated from the current state estimate x_k and error covariance:

$$P_k \chi_0 = x_k, \chi_i = x_k + \sqrt{(n + \lambda)P_k} e_i, \chi_{n+i} = x_k - \sqrt{(n + \lambda)P_k} e_i \quad (65)$$

where, n is the state dimension, $\lambda = \alpha^2(n + \kappa) - n$, e_i are the columns of the identity matrix. α, κ are UKF tuning parameters.

Step 2: Time update (prediction)

Each sigma point is propagated through the nonlinear dynamics:

$$\chi_i^- = f(\chi_i, u) \quad (66)$$

The predicted state and covariance are computed as:

$$P_k^- = \sum_{i=0}^{2n} W_c^i (\chi_i^- - x_k^-)(\chi_i^- - x_k^-)^T + Q, \quad x_k^- = \sum_{i=0}^{2n} W_m^i \chi_i^- \quad (67)$$

where, W_m^i, W_c^i are the weights for the mean and covariance.

Step 3: Measurement update (correction)

The sigma points are passed through the measurement function:

$$Z_I = h(\chi_i^-) \quad (68)$$

The predicted measurement mean and covariance are:

$$z_k^- = \sum_{i=0}^{2n} W_m^i Z, \quad P_{zz}^- = \sum_{i=0}^{2n} W_c^i (Z_i - z_k^-)(Z_i - z_k^-)^T + R \quad (69)$$

Cross-covariance between the state and measurement:

$$P_{xz} = \sum_{i=0}^{2n} W_c^i (\chi_i^- - x_k^-)(Z_i - z_k^-)^T \quad (70)$$

The Kalman gain & state update are:

$$K_k = P_{xz} P_{zz}^{-1}, x_k = x_k^- + K_k (z_k - z_k^-), P_k = P_k^- - K_k P_{zz} K_k^T \quad (71)$$

Step 4: Linearization

The nonlinear system is linearized around the current estimated state:

$$A = \frac{\partial f}{\partial x} \Big|_{x=x_k}, B = \frac{\partial f}{\partial u} \Big|_{x=x_k} \quad (72)$$

The optimal control law is given by:

$$u_k = -K x_k, K = (R_{LQR} + B^T S B)^{-1} B^T S A \quad (73)$$

where S is the solution to the algebraic Riccati equation:

$$A^T S + S A - S B (R_{LQR} + B^T S B)^{-1} B^T S + Q_{LQR} = 0 \quad (74)$$

In the combined UKF-LQR framework:

State Estimation: The UKF provides an estimate $\hat{x}_k$ of the true state.

Control Law: The LQR uses the estimated state to compute the optimal control input:

$$u_k = -K \hat{x}_k \quad (75)$$

The integration of UKF with LQR forms a robust nonlinear LQG controller, suitable for systems like the inverted pendulum. The UKF handles the nonlinear dynamics in state estimation, while the LQR ensures optimal control under uncertainty.

Disturbance rejection in a control system is achieved by minimizing the effect of external disturbances on the system's state.

With Disturbance rejection: The nonlinear dynamics of the inverted pendulum on a cart, subjected to disturbances, are given by:

$$\begin{aligned} \dot{\tilde{x}}(t) &= A \tilde{x}(t) + B u(t) + D d(t) \\ y(t) &= C \tilde{x}(t) + v(t) \end{aligned} \quad (76)$$

where $\tilde{x}(t)$, state deviation from equilibrium. $d(t)$, external disturbance acting on the system. D, disturbance coupling matrix. $v(t)$, measurement noise. $y(t)$, measurement vector.

The Unscented Kalman Filter estimates the system state based on noisy measurements. The LQG controller is designed based on the estimated state $\hat{x}(t)$ provided by the Unscented Kalman Filter (UKF).

The control input is computed as:

$$u(t) = -K \hat{x}(t) \quad (77)$$

Substitute Eq. (77) into Eq. (76),

$$\dot{\hat{x}}(t) = (A - BK) \hat{x}(t) + D d(t) + BK(\tilde{x}(t) - \hat{x}(t)) \quad (78)$$

The term $(\tilde{x}(t) - \hat{x}(t))$ represents the estimation error, which is minimized by the UKF.

The disturbance $d(t)$ propagates through the system via the transfer function:

$$\tilde{x}(s) = G(s) d(s) \quad (79)$$

where,

$$G(s) = (sI - (A - BK))^{-1}D \quad (80)$$

The UKF reduces the estimation error $(\hat{x}(t) - \hat{x}(t))$ by accurately estimating the true state. This ensures the control law $u(t)$ effectively counteracts disturbances:

$$\|(\hat{x}(t) - \hat{x}(t))\| \rightarrow 0 \text{ as } t \rightarrow \infty \quad (81)$$

under the assumption of bounded process and measurement noise.

For bounded disturbances $d(t)$, the closed-loop state deviation satisfies:

$$\|\hat{x}(t)\| \leq \|G(s)\| \|d(t)\| \quad (82)$$

The feedback gain K is chosen to minimize the transfer function $G(s)$, thereby reducing the impact of disturbances. If $G(s)$ has low gain, even large disturbances $d(t)$ will have minimal impact on $\hat{x}(t)$.

Model predictive controller

The utilisation of Model Predictive Control (MPC) increased considerably in recent decades, particularly in industrial processes³⁰. The block diagram is shown in Fig. 5³¹.

Stability is vital for limiting erratic behavior in the regulated system. Stability is critical for preventing erratic behavior in the regulated system³². Model uncertainties may result from errors or changes in the underlying process model, enabling approaches that compensate and minimize these uncertainty³³. The regulating computational complexity is a difficulty, considering the optimization problems connected with MPC can be computationally intensive^{19,34,35}.

$$\bar{x}_{i_{k+1}} = \bar{A}x_{i_k} + \bar{B}u_{i_k} \quad (83)$$

$$\bar{y}_{i_k} = \bar{C}x_{i_k} \quad (84)$$

where

$$x_{i_k} \in R_i^n, \bar{y}_{i_k} \in R_i^o, u_{i_k} \in R_i^g$$

$$\bar{u}_{i_k} = \bar{u}_{i_{k+1}} + \Delta \bar{u}_{i_k} \quad (85)$$

$$\bar{x}_{i_{k+1}} = \begin{bmatrix} \hat{x}_{i_{k+1}} \\ \hat{u}_{i_k} \end{bmatrix} = \hat{A}\bar{x}_k + \hat{B}\bar{u}_k,$$

$$A = \begin{bmatrix} \hat{A} & \hat{B} \\ 0 & I_{i_m} \end{bmatrix}; \quad B = \begin{bmatrix} \hat{B} \\ I_{i_m} \end{bmatrix}; \quad C = [\bar{C} \quad 0] \quad (86)$$

$$J = e_{i_k+N_{i_y}}^T \hat{Q} N e_{K+N_{i_y}} + \sum_0^{N_{i_y}-1} e_{i_k+t}^T \hat{Q} e_{K+t} + \sum_0^{N_{i_u}-1} \Delta u_{i_{K+t}}^T \hat{R} \Delta u_{i_{K+t}} \quad (87)$$

$$e_k = y_{i_k} - u_{i_k}$$

N_{i_y} = prediction horizon, N_{i_u} = control horizon, W = weighting matrix used in cost function.

$$N_{i_u} \leq N_{i_y}; \quad \bar{Q} = \bar{Q}^T \geq 0; \quad \bar{R} = \bar{R}^T \geq 0;$$

$$y = \alpha x_{i_k} + \phi \hat{u}_d \quad (88)$$

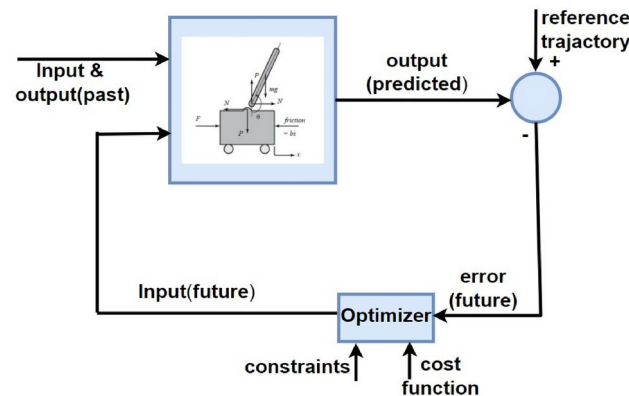


Fig. 5. Block diagram of MPC with IP system.

$$Y = \begin{bmatrix} y_{i_k+1} \\ y_{i_k+2} \\ \vdots \\ y_{i_k+N} \end{bmatrix}, \quad W = \begin{bmatrix} w_{i_k+1} \\ w_{i_k+2} \\ \vdots \\ w_{i_k+N} \end{bmatrix}, \quad \hat{u}_d = \begin{bmatrix} \Delta u_{i_k} \\ \Delta u_{i_k+1} \\ \vdots \\ \Delta u_{i_k+N-1} \end{bmatrix} \quad (89)$$

$$J = J_1 + (Y - W)^T \hat{Q} (Y - W) + \hat{u}_d^T \bar{R} \hat{u}_d \quad (90)$$

$$J_1 = (y_{i_k} - w_{i_k})^T \hat{Q} (y_{i_k} - w_{i_k})$$

$$\bar{R} = \text{diag}(R_{i_K}); R_{i_K} = \bar{R}$$

$$\hat{Q} = \text{diag}(Q_{i_K}); Q_{i_K} = \hat{Q}$$

$$\alpha = \begin{bmatrix} CA \\ CA^2 \\ \vdots \\ CA^N \end{bmatrix} = \begin{bmatrix} \bar{CA} & \bar{CB} \\ \bar{CA}^2 & \bar{C}(\bar{A} + I_n)\bar{B} \\ \vdots & \vdots \\ \bar{CA}^N & \bar{C}(\sum_{i=0}^{N-1} \bar{A}^i)\bar{B} \end{bmatrix} = [\alpha_1 \quad \alpha_2] \quad (91)$$

$$\phi = \begin{bmatrix} CB & 0 & \vdots & \vdots & \vdots & 0 & 0 \\ CAB & CB & \vdots & \vdots & \vdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ CA^{N-1}B & CA^{N-2}B & \vdots & \vdots & \vdots & CAB & CB \end{bmatrix} \quad (92)$$

Implementation of the methodology to the system

In this section, methodology discussed in above section has been implemented to system based on the inverted pendulum on a cart for the performance analysis¹⁶. The parameter that is used for Inverted Pendulum is shown in Table 2^{12–16}. As application of its has vital impact on robotic system for balancing the object on motion. Objective function present in Eq. (32) indicates significance of R. Simulation results for varying values of in the LQR controller are provided in Fig. 6a, b to facilitate a comparative analysis of system performance. The value of R directly impacts the control effort, revealing the trade-offs between control energy and stability. By

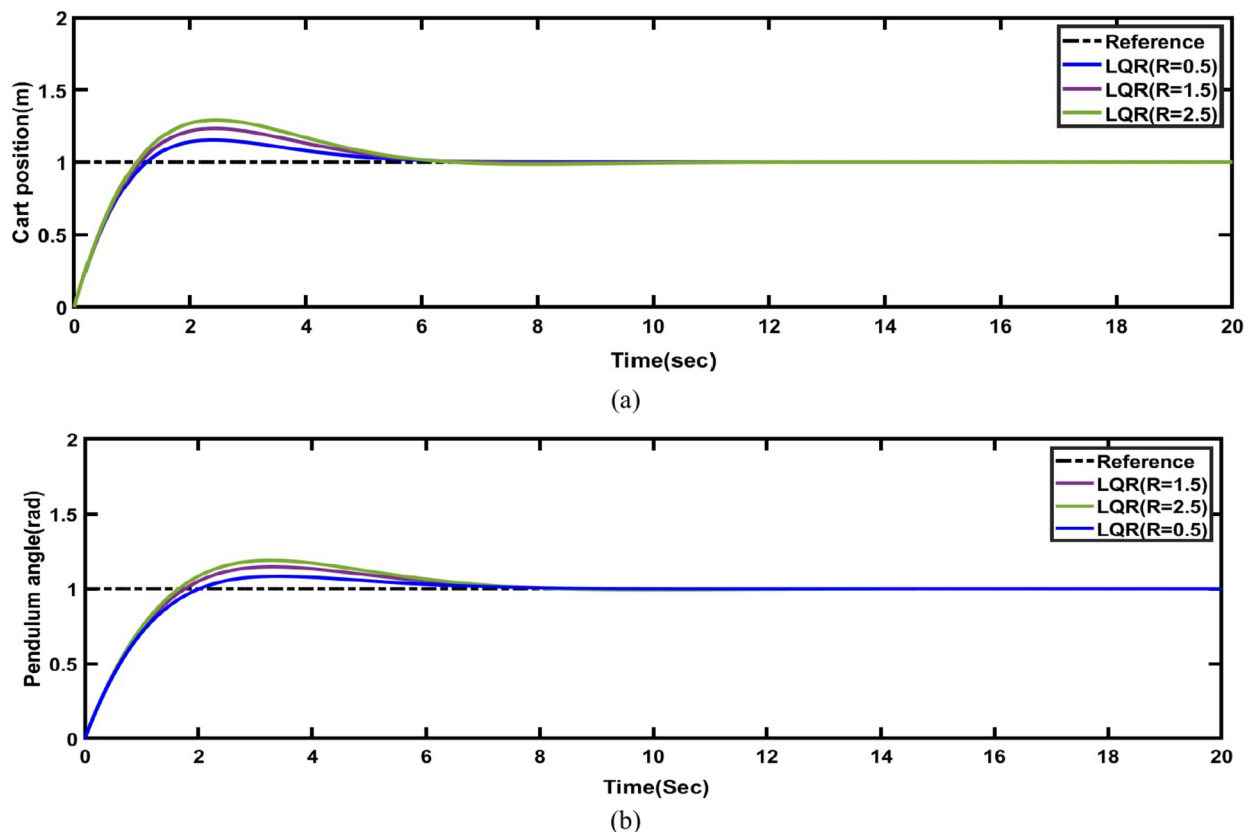


Fig. 6. The overall performance of system (LQR).

examining the cost function for each R , the analysis highlights how different values influence the balance between performance and energy efficiency. performance of LQR is based on minimization of on the feedback control law that minimizes the cost.

$$u(t) = -R^{-1}B^T P x(t) - K x(t) \quad (93)$$

This equation shows that variation of R have been responsible for driving changes on the regulated output performance. Value of R has been varied with three cases $R = [0.5 \ 1.5 \ 2.5]$. The performance analysis of inverted pendulum with change in value of R has been collected based on the output For LQR Controller as cart position and pendulum angle presented in Fig. 6a and b respectively. To maintain the control effort low i.e. $[0.3858, 0.5827, 0.7432]$ for Eq. (32), R must be selected positive small value as 0.5, as it give the faster response compared to the $R = 1.5$ and 2.5. and for the LQR Controller the value of J is 0.784 for Eq. (33). Then. It has been compared with Classical PID, LQG and MPC presented in Fig. 7.

As per the Eqs. (44) and (48) estimation error covariance matrix and kalman gain of LQG are as follows:

$$P = \begin{bmatrix} 0.0169 & -0.0012 & 0.0038 & -0.0085 \\ -0.0012 & 0.1062 & -0.0004 & 2.1141 \\ 0.0038 & -0.0004 & 0.0996 & -0.0006 \\ -0.0085 & 2.1141 & -0.0006 & 46.5016 \end{bmatrix}; \text{ the kalman gain is } K = \begin{bmatrix} 0.7525 & 0.0026 \\ -0.0588 & 0.5252 \\ 0.0026 & 0.1883 \\ -0.1455 & -0.0006 \end{bmatrix}.$$

When LQG integrated with UKF, from Eq. (68), the value of u_k will be -0.1254 . and the value of $\hat{x}_k$ is $[-0.05 \ 0.02 \ 0.14 \ 0.01]$ and initial covariance matrix p_k and kalman gain of LQG with UKF are:

$$P_k = \begin{bmatrix} 0.0752 & -0.0059 & 0.0003 & -0.0146 \\ -0.0059 & 0.2940 & 0.0525 & 0.0013 \\ 0.0003 & 0.5225 & 0.0188 & -0.0001 \\ -0.0146 & 0.0013 & -0.0001 & 0.0132 \end{bmatrix}; K = \begin{bmatrix} 0.5025 & 0.0018 \\ -0.0241 & 0.3456 \\ 0.0015 & 0.1023 \\ -0.0553 & -0.0001 \end{bmatrix}.$$

Open loop responses of LQG and LQG With UKF has been shown in Fig. 8a, b.

This technique has also been compared with Model Predictive Control (MPC), which requires multiple coefficients that accurately represent system response. and open loop response is shown in Fig. 8a, and b. Based on the collected data for each technique, LQG With UKF yields a faster response for both cart position and pendulum angle, with rise times of 1.30 s and 1.24 s, respectively. It achieves faster settling in just 4.1 s and 3.9 s.

The system has been subjected to changes in input in order to assess the resilience of various techniques. The system performance with uniform random repeating sequence staircase as input function $u(t) = [0.02, 0.01, 0.06]$

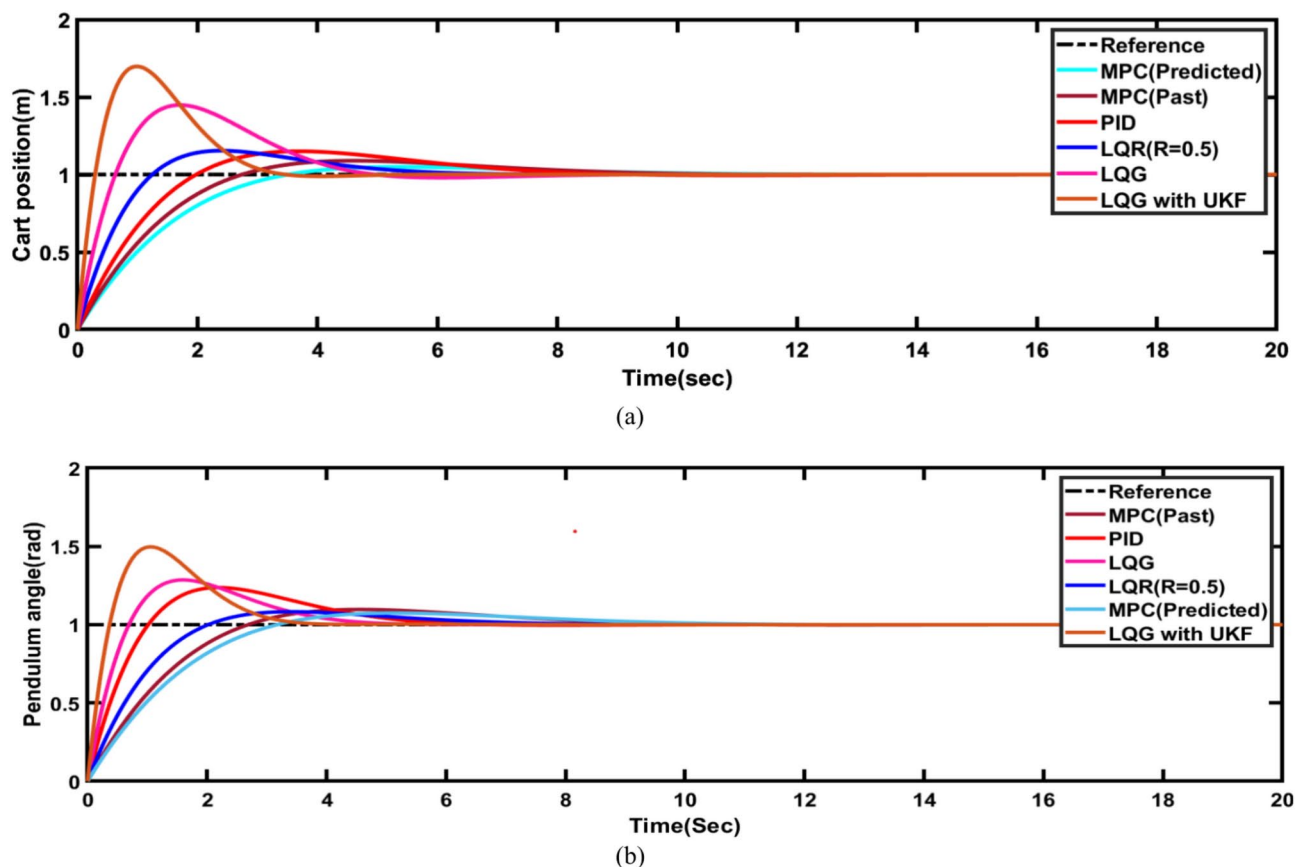


Fig. 7. Performance comparison between the controllers.

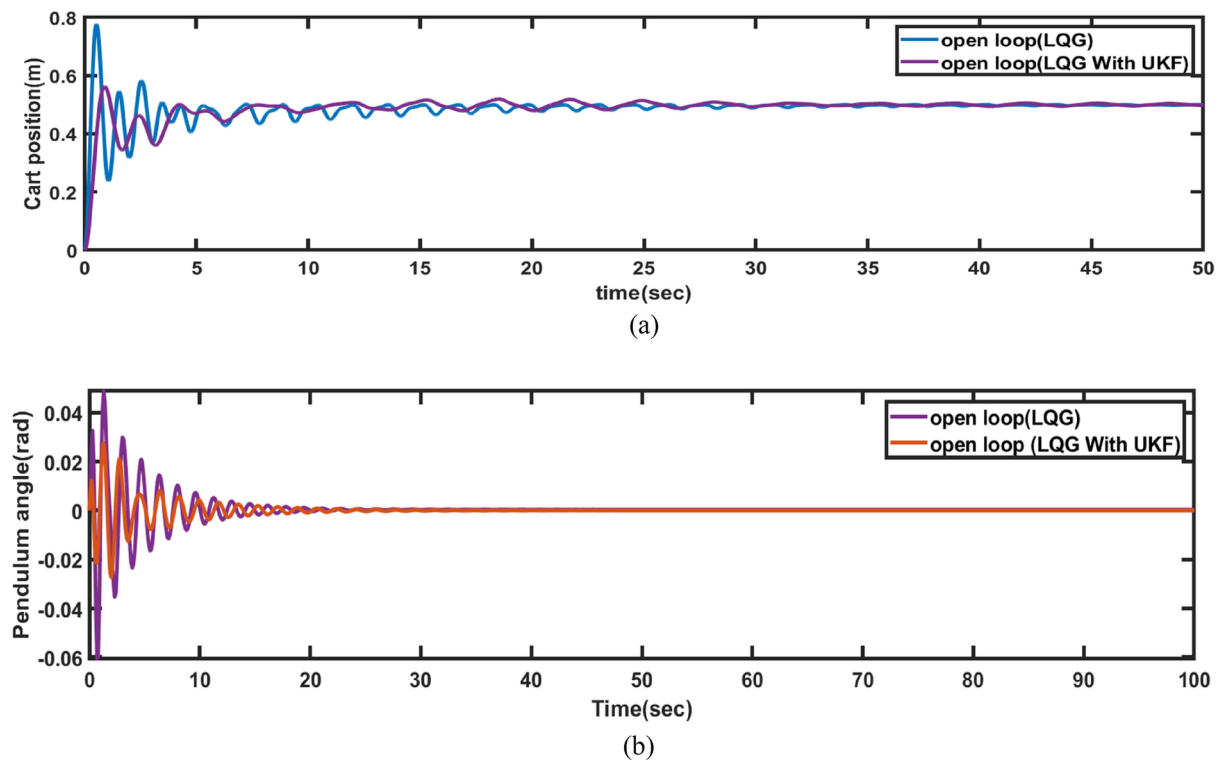


Fig. 8. (a) Response of Open-loop. (b) Response of Open-loop.

applies to the system and performance based on change in input has been collected presented in Fig. 9a, b and the statistical information has been collected in Table 4. It is verified that LQG with UKF is faster and overall performance is more effective than others when an inverted pendulum is mounted on a cart.

To verify the robustness of the defined methodology, the unforced system undergoes disturbance signal, which is shown in Fig. 10, Based on the sharp, localized deviation around 10 s, performance analysis of the disturbance is most likely an impulse disturbance, which indicates that the application of LQG With UKF is much better as compared to other techniques.

Conclusion

This study addresses a significant challenge in control engineering: the control of a nonlinear system, specifically an inverted pendulum system on a cart. The intricacy of this problem arises from the interactions involved between highly coupled states and nonlinear components within the system. The linearization method based on the vertical pendulum position was used to estimate the system dynamics in order to deal with this complexity. The linear model has been meticulously crafted to guarantee complete controllability and observability.

Based on a linear model, the system is subjected to various methodologies, including traditional PID, LQR, LQG, and MPC, each with a distinct collection of environmental variables. Each technique has been extensively explained, along with the functionality of the system. The graphical and statistical responses illustrate LQG With UKF's superiority compared to alternative methods. The LQG controller works well for inverted pendulum systems that require optimal control and state estimation. The limitation of the LQG controller is the need to precisely integrate the LQR and Kalman filters. The integration of the Unscented Kalman Filter (UKF) with the LQG controller significantly enhances its performance, providing a robust solution for inverted pendulum systems by effectively addressing state estimation and control challenges, particularly in the presence of nonlinearities. This provides a comprehensive framework for addressing every aspect of system estimation, control, and uncertainty with a cohesive approach. In order to improve a posterior estimate by adding a new measurement to a prior estimate, the measurement update equations are to provide feedback.

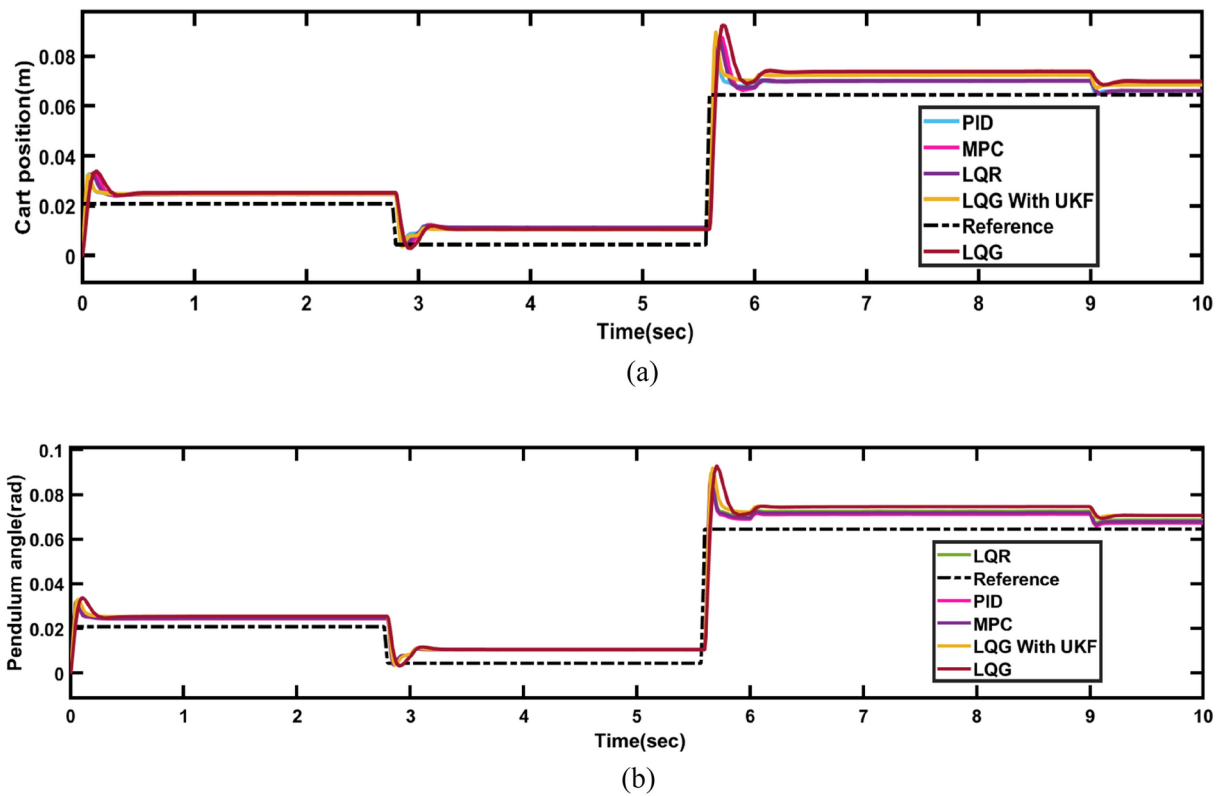


Fig. 9. (a) Performance comparison between the controllers with staircase input. (b) Performance comparison between the controllers with staircase input.

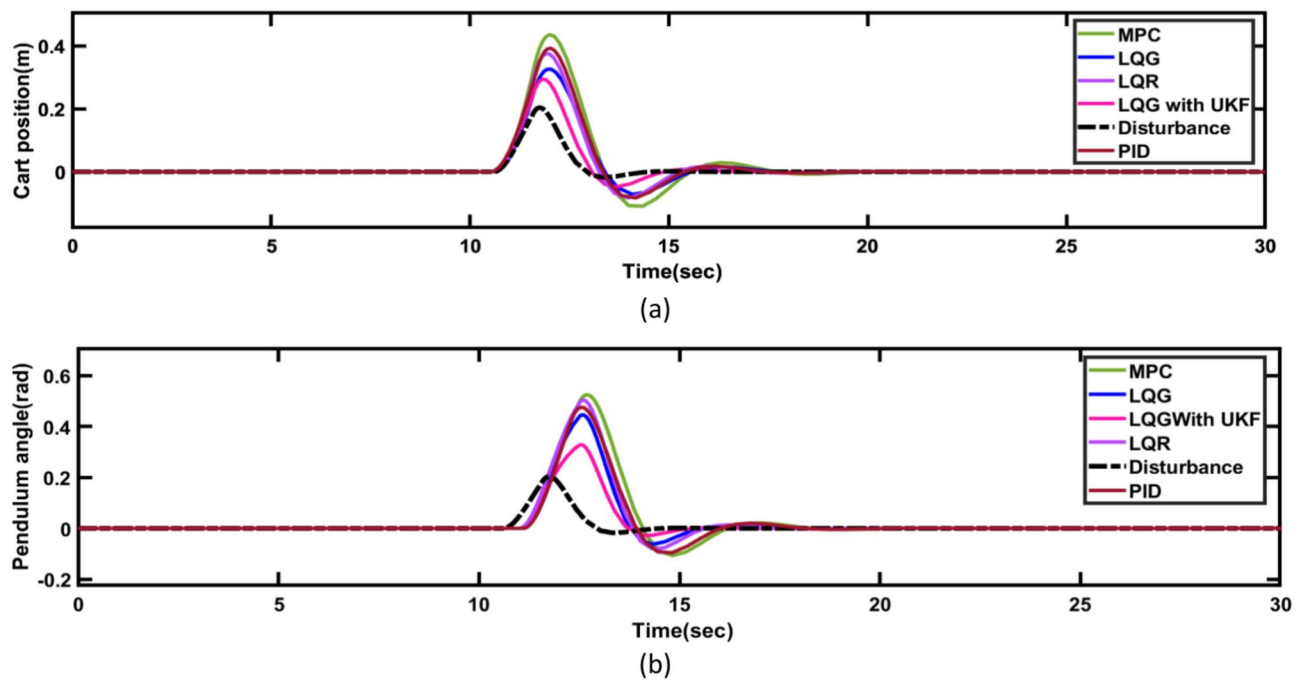


Fig. 10. Disturbance tracking of inverted pendulum.

Methodology	Parameter	Rise time(sec)	Settling time(sec)	Max. overshoot	Max	Min
PID	Cart position	1.62	6.4	0.32	1.15	0
	Pendulum angle	1.26	6.6	0.31	1.237	0
LQR(R = 0.5)	Cart position	1.42	5.9	0.35	1.154	0
	Pendulum angle	1.54	7.1	0.38	1.083	0
LQR(R = 1.5)	Cart position	1.48	6.1	0.37	1.234	0
	Pendulum angle	1.57	7.3	0.40	1.146	0
LQR(R = 2.5)	Cart position	1.52	6.3	0.40	1.292	0
	Pendulum angle	1.59	7.5	0.43	1.189	0
LQG	Cart position	1.05	4.5	0.30	1.448	0
	Pendulum angle	1.08	4.3	0.25	1.286	0
LQG with UKF	Cart position	1.30	4.1	0.70	1.712	0
	Pendulum angle	1.24	3.9	0.49	1.405	0
MPC(Past)	Cart position	1.80	7.3	0.55	1.089	0
	Pendulum angle	1.86	7.5	0.53	1.097	0
MPC(Predicted)	Cart position	2.22	7.8	0.58	1.049	0
	Pendulum angle	1.98	8.0	0.55	1.074	0

Table 3. Position and angle response for inverted pendulum on a cart (analytical data from Fig. 7a, b).

Methodology	Parameter	Rise time (s)	Settling time (s)	Max. overshoot	Max.	Min.
PID	Cart position	0.0320	9.112	0.95%	0.0874	0
	Pendulum angle	0.0015	9.120	0.90%	0.0825	0
LQR (R = 0.5)	Cart position	0.0375	9.199	0.86%	0.0875	0
	Pendulum angle	0.0025	9.207	0.75%	0.0835	0
LQG	Cart position	0.0310	9.305	0.55%	0.0872	0
	Pendulum angle	0.0030	9.311	0.50%	0.0865	0
LQG With UKF	Cart position	0.0305	9.104	0.50%	0.0862	0
	Pendulum angle	0.0025	9.110	0.45%	0.0855	0
MPC	Cart position	0.3390	9.204	0.62%	0.08965	0
	Pendulum angle	0.0020	9.210	0.55%	0.0827	0

Table 4. Position and angle response for inverted pendulum on a cart for stair sequence input.

Data availability

The datasets generated during this study are not publicly available due to the following reasons: The datasets that are used for system parameters in Table 2 are available from the reference¹⁶, and we generated all the tables from this system parameter data on our own in a Matlab environment. And all the data generated in Tables 3 or 4 are not publicly available but are available from the corresponding author on reasonable request.

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Author contributions

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Competing interests

The authors declare no competing interests.

Additional information

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